

Project Planning Phase

Project Planning (Product Backlog, Sprint Planning, Stories, Story Points)

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and

what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

As an individual contributor, Divya Kauluri was assigned to every task below across all 3 sprints.

Product Backlog & Sprint Schedule

Epic	Task	Points	Priority	Sprint
Data Collection	Download and verify official dataset (345K rows, 1997–2021)	2	High	Sprint 1
Data Collection	Confirm dataset columns: State, District, Crop, Year, Season, Area, Production	1	High	Sprint 1
Data Preparation	Clean dataset using Python Pandas (rename, strip whitespace, extract year, add Yield)	3	High	Sprint 1
Data Preparation	Connect cleaned CSV to Tableau; set State Name geographic role	2	High	Sprint 1
Data Visualization	Build State-wise Production filled map with Season + Year filters	3	High	Sprint 1
Data Visualization	Build Year-wise Total Production and Trend of Agricultural Yield line charts (1997–2021)	2	High	Sprint 1
Data Visualization	Build KPI Summary panel and Leading States by Production chart	2	High	Sprint 2
Data Visualization	Build Season-wise Production chart (pie) and Production Distribution by Unit Type chart (donut)	2	Medium	Sprint 2
Data Visualization	Build Cultivated Area by Crop and State-wise Cultivated Area Comparison charts (exclude Coconut outlier)	3	Medium	Sprint 2
Dashboard	Assemble Farmer's View dashboard	3	High	Sprint 2
Dashboard	Assemble Policy Analyst's View dashboard	3	High	Sprint 2
Dashboard	Assemble Investor's View dashboard	2	Medium	Sprint 2
Story	Build Tableau Story with story points and captions	3	Medium	Sprint 3
Performance Testing	Test all filters for responsiveness; document load time and data volume	2	Medium	Sprint 3
Web Integration	Publish workbook to Tableau Public; verify URL access	2	High	Sprint 3
Documentation	Complete all GitHub repository deliverables (docs, report, screenshots)	5	High	Sprint 3
Documentation	Record demonstration video (60–90 seconds)	3	High	Sprint 3

Velocity

Sprint 1 = 13 points | Sprint 2 = 16 points | Sprint 3 = 12 points | Total = 41 story points. Sprint 2 carried the heaviest load because it covered both dashboard assembly and the Tableau Story, the two most integration-heavy

deliverables; Sprint 3's lighter total reflects testing and documentation work, a normal taper on a single-contributor schedule.